MTH129 Introduction to AI for Mathematical Sciences (MS)

2025-2026 Academic Year
Semester 2

Activation Functions

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Activation Functions

Here, we summarize the pros and cons of several activation functions, including ReLU, Leaky ReLU, Softplus, and Sigmoid.

ReLU (Rectified Linear Unit) and Advantages

- $\sigma(x) = \max(0, x) = \frac{x+|x|}{2}$
- Range: $[0, +\infty)$
- **Simple Computation:** It only requires comparison and zeroing operations, without exponentiation, multiplication or division, enabling fast training and inference.
- **Mitigates Vanishing Gradients:** In the positive region ($x > 0$), the gradient is constantly 1, effectively propagating error signals through deep networks.
- **Sparse Activation:** Negative outputs are zeroed, leaving some neurons inactive and enhancing network sparsity and computational efficiency.
- **Easier Optimization:** Compared to Sigmoid/Tanh, its linear portion enables faster convergence.

Disadvantages of ReLU

- **Dying Neurons:** If inputs remain negative for extended periods, gradients become zero, halting weight updates and permanently deactivating neurons. This phenomenon is known as the "dying ReLU" problem.
- **Sensitivity to Outliers:** No upper bound in the positive region allows extremely large inputs to cause exploding activations.
- **Non-Differentiable at Zero:** While implementations typically assign a derivative of 0 or 1 at $x = 0$, it is theoretically not fully smooth.
- **Non-Zero Mean:** Outputs are always non-negative, causing biased inputs to the next layer and potentially slowing gradient descent.

Leaky ReLU

- $\sigma(x) = \begin{cases} x & \text{for } x \geq 0 \\ \alpha x & \text{otherwise} \end{cases}$ with a small α like 0.01
- Range: $(-\infty, +\infty)$
- **Linearity in the Positive Region:** When $x > 0$, the gradient remains 1, fully retaining benefits such as **simple computation**, **no vanishing gradients**, and **fast convergence**.
- **Approximate Sparsity:** Outputs in the negative region are scaled to a small value αx (close to 0 but not exactly 0). While not fully sparse, this still suppresses most noisy activations in practice, preserving some degree of sparsity.
- **No "Dying ReLU" Problem:** Leaky ReLU introduces a small constant non-zero gradient (α) in the negative region, ensuring weights continue receiving update signals even with prolonged negative inputs. This allows neurons to potentially recover and return to the positive region, avoiding permanent death.

Softplus

- **Softplus**$(x) = \log(1 + e^x)$
- Range: $(0, +\infty)$
- **Smooth:** Unlike ReLU, which is non-differentiable at 0, Softplus is infinitely differentiable. This makes it more compatible with gradient-based optimization, particularly in scenarios that require second-order derivatives or the Hessian matrix.
- **No Vanishing Gradient Problem on the Positive Axis:** As $x \to \infty$, **Softplus**$(x) \approx x$, and the gradient approaches 1. This prevents the vanishing gradient problem seen in Sigmoid or tanh at their saturation points, facilitating better gradient flow in deep networks.
- **No "Dying ReLU" Problem:** For $x < 0$, Softplus outputs a small positive value (when $x = -10$, the output is approximately 0.000045), rather than being completely zero like ReLU.

Disadvantages of Softplus

- **Vanishing Gradient on the Negative Axis:** Although not as severe as with Sigmoid, the gradient of Softplus approaches 0 as $x \to -\infty$. This implies that neurons in the negative region of a deep network still contribute a negligible gradient, which is less effective than functions like **Leaky ReLU** that maintain a fixed small slope in the negative region.
- **Higher Computational Cost than ReLU:** Softplus involves exponential and logarithmic operations, making it slower than ReLU. For extremely large-scale networks (such as GPT-4), even a minor overhead can accumulate significantly.
- **Non-Zero Mean**

Sigmoid

- **Sigmoid**$(x) = \sigma(x) = \dfrac{1}{1 + e^{-x}}$
- Range: $(0, 1)$
- **Output Range** $(0, 1)$**, Interpretable as Probability:** This is the greatest advantage of Sigmoid. With output values strictly between 0 and 1, it is naturally suitable for representing the positive class probability in binary classification problems.
- **Smooth**
- **Simple Derivative Form:** The derivative is given by $\sigma'(x) = \sigma(x)(1 - \sigma(x))$. This allows the gradient to be computed quickly using the function's own output, without extra computation.
- **Center Symmetry (After Transformation):** Although $\sigma(x)$ itself is not zero-centered, the transformation $2\sigma(x) - 1$ yields the $\tanh$ function (with a range $(-1, 1)$), which is zero-centered.

Disadvantages of Sigmoid

- **Vanishing Gradient:** When $|x|$ is large (e.g., $|x| > 5$), the output of Sigmoid approaches 1 or 0, and its derivative approaches 0.
- **Higher Computational Cost**
- **Non-Zero Mean**